# Apple Diseases

Identification, causes, favourable conditions, management and verified remedies

Nine major diseases with a focus on the Himalayan apple belt (Jammu & Kashmir, Himachal Pradesh, Uttarakhand)

Primary sources: CIB&RC (Insecticides Act, 1968) Major Uses of Pesticides, ICAR journals and ICAR-CITH, Dr YS Parmar UHF Nauni, National Horticulture Board, and university extension material for pathogen biology. Compiled 18 October 2026.

# Contents

# About this document

## Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. Disease descriptions come from ICAR and state agricultural university publications, NHB, and university extension pages for pathogen biology. Chemical recommendations are included only where they could be matched to the CIB&RC compilation (up to 30/09/2025) of registered uses on apple or to a published ICAR / state agricultural university / NHB recommendation, and each is tagged to show how it was verified.

## Verification tags used in the chemical tables

**V1 -** Verified against CIB&RC. The product is listed for APPLE and this disease/claim in the CIB&RC 'Major Uses of Pesticides - Fungicides' compilation up to 30/09/2025. Doses are reproduced as printed; for fruit trees CIB&RC gives a spray-solution concentration and a volume per tree or per ha.

**V2 -** Published trial or recommendation only. Taken from an ICAR / Dr YS Parmar UHF Nauni publication or from the National Horticulture Board text (citing ICAR). No matching apple label claim was found in the 2025 CIB&RC list, so confirm the current registration and label before use.

**-- -** Not recommended / no chemical option verified. The sources reviewed do not support a chemical recommendation, so none is given.

## Important cautions

- The CIB&RC compilation used for V1 is dated up to 30/09/2025 and was read from a copy hosted on a university website, not from the live register. Registrations, label claims and permitted uses change. Always check the current label and the live register (cibrc.gov.in) and follow the pre-harvest interval (PHI, in days) printed on the label.
- For trees, CIB&RC gives doses as a concentration (% or g/ml per 100 L or per litre) and a spray volume per tree (typically 10 L/tree for apple); per-hectare figures are used where CIB&RC gives them. In a few combination entries the unit printed beside the dose is unclear; those rows say 'as printed'. Spray volume must match tree size and sprayer.
- Rotate fungicide groups across sprays. Apple scab populations with resistance to benzimidazoles, dodine, strobilurins and DMI fungicides have been reported elsewhere (VER); alternating or mixing groups and limiting applications per group is the recommended strategy.
- Apple is grown mainly in Jammu & Kashmir, Himachal Pradesh and Uttarakhand (VER, WR1), so regional notes refer to these states. No state-wise spray calendar from SKUAST-Kashmir, YSPUHF Nauni or the HP Horticulture Department was retrieved, so none is reproduced beyond the ICAR schedule published by NHB.
- Cross-check against the 2012 compilation: bitertanol, fenarimol and flusilazole 40% EC (scab) and dinocap (powdery mildew) were listed for apple in 2012 but were not found for apple in the 2025 list, so they were removed from the tables.

# Summary table

| Disease | Pathogen / type | Main non-chemical measures | Chemical status |
|---|---|---|---|
| Apple Scab | Venturia inaequalis (asexual stage Spilocaea pomi) (Fungal (leaf-litter-borne, rain-splashed)) | Time sprays to infection periods; fallen-leaf management; rotate fungicide groups | Many V1 options (protectants, triazoles and combinations); V2 trial carbendazim + flusilazole |
| Marssonina Leaf Blotch (Premature Leaf Fall) | Marssonina coronaria (sexual stage Diplocarpon mali) (Fungal (leaf-litter-borne)) | Destroy fallen leaves; preventive spraying before mid-June | V1 options now listed (kresoxim-methyl, mancozeb WG, dodine, tebuconazole and combinations); copper hydroxide V2 |
| Powdery Mildew | Podosphaera leucotricha (Fungal (obligate, bud-borne)) | Prune out infected shoots; open canopy; extra care in nurseries | V1: sulphur, lime sulphur, aureofungin, bupirimate, metrafenone and combinations; V2 from NHB |
| Collar Rot | Phytophthora cactorum (Oomycete (soil-borne)) | Drainage, resistant rootstock on wet sites, solarization, oil cakes | None verified (tag --) |
| White Root Rot | Dematophora necatrix (Rosellinia necatrix) (Fungal (soil-borne)) | Paron rootstock, nursery-site rotation, intercrops, solarization | V1: aureofungin only; carbendazim not included |
| Alternaria Leaf Blotch | Alternaria alternata f. sp. mali (Alternaria mali); other Alternaria spp. also associated (Fungal (leaf-litter-borne, toxin-producing)) | Manage leaf litter; preventive sprays; no resistant cultivar found | V1: dodine, hexaconazole, mancozeb WG and combinations |
| Sooty Blotch and Flyspeck | Fungal complex: Peltaster fructicola, Geastrumia polystigmatus, Leptodontium elatus (sooty blotch); Zygophiala jamaicensis (flyspeck) (Fungal (surface blemish; twig-borne)) | Canopy aeration and drying; scab-programme fungicides | V1: mancozeb, captan (shared with scab) |
| Fire Blight | Erwinia amylovora (Bacterial (blossom- and shoot-infecting)) | Burn affected trees and host plants; protect spring blossom | V1: streptomycin + tetracycline at bloom |
| Cedar Apple Rust | Gymnosporangium juniperi-virginianae (Fungal rust (needs two hosts: juniper and apple)) | Keep apple away from juniper; remove galls in early spring (no Indian report found) | None (no apple rust claim in CIB&RC 2025; tag --) |

Quick separation of look-alikes: bright orange leaf spots near junipers point to cedar apple rust; golden bacterial ooze, drooping shoot tips without browning and a scorched look point to fire blight; olive-green spots, cracked scabby fruit and spring onset point to scab; dark brown blotches on the upper leaf surface from mid-June followed by yellowing and leaf fall point to Marssonina blotch; a white powdery coating on shoots and leaves points to powdery mildew; decline with a girdled collar and reddish inner bark at the soil line points to collar rot; wilting, leaf bronzing and a white mycelial mat on dark roots point to white root rot; reddish-brown leaf spots with purplish borders (and petiole lesions) point to Alternaria blotch; superficial smudges and black specks on the fruit skin with no decay point to sooty blotch and flyspeck.

# 1. Apple Scab

Apple | Venturia inaequalis (asexual stage Spilocaea pomi) | Fungal (leaf-litter-borne, rain-splashed)

**At a glance.** The most dreaded foliar and fruit disease of apple in the north-western Himalayas. Cool, wet spring weather drives infection. Control rests on timed protectant and systemic fungicide sprays, with rotation of groups.

## 1. Identification

**Symptoms**

- Leaves: scattered, circular brown or olive-green spots, noted on the undersurface of leaves borne on fruit spurs. Large lesions cause yellowing, defoliation and fruit drop (NHB).
- Fruit: spots first develop around the calyx (blossom) end, later anywhere on the fruit. Scabbed areas crack, letting other pathogens in and causing rots in the field or in storage (NHB).
- Primary (ascospore) infections are usually limited to one or two spots per leaf; secondary (conidial) infections are far more numerous and can cover the whole leaf surface (VER).

**Distinguishing features**

- Marssonina blotch gives dark brown blotches on the UPPER leaf surface from about the second week of June (BLO); powdery mildew shows a white powdery coating (NHB).
- Growers and traders have near-zero tolerance for scabbed fruit, so even light infection reduces marketable yield (VER).

## 2. Causes and progression

- Pathogen: Venturia inaequalis (Cooke) Wint. Asexual stage (Spilocaea pomi) lives on living plant parts; the sexual (perithecial) stage develops on dead fallen leaves (VER).
- Primary inoculum: ascospores from overwintered leaves. In Himachal Pradesh mature ascospores were present at silver tip (Mar-Apr), with peak maturity at petal fall (Apr-May); the average ascospore emission period was 74 days (FOR).
- Progression: scab appearance corresponded with the first infection period after bud break; secondary conidial cycles then multiply infection through spring and early summer (FOR, VER).
- Symptom delay: a light infection period needed at least 3 extra days for symptoms to appear (leaf wetness 9.0-13.4 h, mean temperature 8.5-21 C); moderate and severe periods (leaf wetness over 15 h) needed about 1 extra day (MIL).
- Economic loss: up to 70 % of production value from fruit and pedicel infection with poor control, and up to 100 % with no control (VER).

## 3. Favourable conditions

- Cool, moist climate in early spring; endemic where humidity and rainfall are high in spring and early summer; the pathogen perpetuates under shady, moist conditions (VER).
- Infection depends on hours of leaf wetness and temperature (Mills infection periods); an average of 9 infection periods was observed in spring and early summer in the Kotkhai belt of Himachal Pradesh (FOR, MIL).
- Prolonged wet spells later in the season can cause a secondary spread, as reported in South Kashmir orchards (GK; press report).

## 4. Prevention and management

**Cultural practices, sanitation, resistant varieties and preventive measures**

- Time sprays to weather: Mills infection-period monitoring helped reschedule fungicide applications in Himachal Pradesh (MIL, FOR).
- Because the sexual stage forms on fallen leaves (VER), the ICAR schedule published by NHB includes a urea spray (5 kg per 100 L) before leaf fall; the NHB text does not state its purpose.
- Resistance management: restrict the number of applications per fungicide class and mix or alternate classes (VER).
- Resistance to dodine, kresoxim-methyl, myclobutanil, thiophanate-methyl, DMI and benzimidazole fungicides has been reported in scab populations in other countries (VER).

**Regional note (Himalayan apple belt)**

Apple scab is the most dreaded disease on commercial apple cultivars in the Kashmir valley and a major problem in Himachal Pradesh (VER, GK). Follow the state horticulture department and KVK spray advisory for your orchard's elevation and stage.

## 5. Remedies and treatment

### Chemical options with verification status

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Mancozeb 75% WP | Scab & sooty blotch | 30 g per tree in 10 L water | 10 L/tree | V1 | CIB&RC 2025; PHI not stated |
| Mancozeb 75% WG | Scab, premature leaf fall, Alternaria spot/blight, sooty blotch | 25-30 g per tree | 10 L/tree | V1 | CIB&RC 2025; PHI 40 d |
| Captan 50% WP | Scab | 2.5 kg/ha | 750-1000 L | V1 | CIB&RC 2025; PHI not stated |
| Captan 75% WP | Scab, fly speck, bitter rot | 0.12% a.i. (1667 g formulation per 1000 L) | 15-20 L/tree | V1 | CIB&RC 2025; PHI 8 d. Footnote warns of possible 'resetting' (as printed; probably russeting) on Red Delicious |
| Chlorothalonil 75% WP | Scab | 0.2% formulation (200 g/100 L); use from dormancy to green tip | 10 L/tree | V1 | CIB&RC 2025; PHI 45 d |
| Propineb 70% WP | Scab | 0.30% (300 g/100 L) | as required | V1 | CIB&RC 2025; PHI 30 d |
| Metiram 70% WG | Scab | 0.3% (300 g/100 L) | as required | V1 | CIB&RC 2025; PHI 21 d |
| Zineb 75% WP | Scab, black rot | 1.5-2.0 kg/ha | 750-1000 L | V1 | CIB&RC 2025; PHI not stated |
| Ziram 80% WP | Scab | 1.5-2.0 kg/ha | 750-1000 L | V1 | CIB&RC 2025; PHI 21 d |
| Dithianon 75% WP | Scab | 1800 g/ha | 2400 L | V1 | CIB&RC 2025; PHI 14-21 d |
| Dodine 65% WP | Scab | 0.075% formulation | 10 L/tree | V1 | CIB&RC 2025; PHI 21 d |
| Sulphur 80% WG | Scab | 1.875-2.5 kg/ha | 750-1000 L | V1 | CIB&RC 2025; PHI not stated |
| Kresoxim-methyl 44.3% SC | Apple blotch, leaf scab and fruit scab | 0.05% formulation (50 ml/100 L) | per tree size | V1 | CIB&RC 2025; PHI 7 d |
| Difenoconazole 25% EC | Scab | 0.015% (15 ml/100 L) | as required | V1 | CIB&RC 2025; PHI 14 d; HP field trial at 0.015% (VER) |
| Hexaconazole 5% EC | Scab | 0.05% (50 ml/100 L) | as required | V1 | CIB&RC 2025; PHI 30 d; HP field trial at 0.05% (VER) |
| Myclobutanil 10% WP | Scab | 0.04% | 10 L/tree | V1 | CIB&RC 2025; PHI 21 d |
| Penconazole 10% EC | Scab | 0.05% (50 ml/100 L) | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Carbendazim 50% WP | Scab | 2.5 g formulation per 10 L water per tree (0.025%) | 10 L/tree | V1 | CIB&RC 2025; PHI not stated. Weakest in HP trial (about 67 % control) (VER) |
| Thiophanate-methyl 70% WP | Scab | 715 g/ha | 750-1000 L | V1 | CIB&RC 2025; PHI 3 d |
| Captan 70% + Hexaconazole 5% WP | Scab and Alternaria blotch | 0.06-0.1% (6-10 g per 10 L) | per tree size | V1 | CIB&RC 2025; PHI 10 d |
| Carbendazim 12% + Mancozeb 63% WP | Fruit scab & powdery mildew | 0.25% formulation | per canopy | V1 | CIB&RC 2025; PHI 20 d |
| Chlorothalonil 40% + Difenoconazole 4% SC | Powdery mildew, premature leaf fall and scab | 800 (ml/g) per ha | 1000 L | V1 | CIB&RC 2025; PHI 24 d |
| Difenoconazole 4% + Propineb 64% DF | Scab, powdery mildew, premature leaf fall | 0.125% formulation (125 g/100 L) | max 1000 L/ha | V1 | CIB&RC 2025; PHI 24 d |

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| | (Marssonina coronaria) | | | | |
| Fluopyram 21.37% + Trifloxystrobin 21.37% SC | Marssonina blotch, scab, powdery mildew | 50-60 ml per 100 L | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Fluxapyroxad 75 g/l + Difenoconazole 50 g/l SC | Scab and powdery mildew | 1000 ml/ha | 2500 L | V1 | CIB&RC 2025; PHI 33 d |
| Hexaconazole 4% + Zineb 68% WP | Scab, premature leaf fall, Alternaria, powdery mildew, core rot | 0.25% (25 g per 10 L) | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Tebuconazole 6.7% + Captan 26.9% SC | Powdery mildew, Alternaria leaf spot/blight, scab | 0.25% (2.5 ml/L) | 10 L/tree | V1 | CIB&RC 2025; PHI 10 d |
| Azoxystrobin 11% + Tebuconazole 18.3% SC | Scab, powdery mildew & premature leaf fall | 1.0 (per litre, as printed) | 8-12 L/tree | V1 | CIB&RC 2025; PHI 10 d |
| Trifloxystrobin 10% + Difenoconazole 12.5% + Sulphur 3% SC | Scab | 450 ml/ha | 1000 L | V1 | CIB&RC 2025; PHI 7 d |
| Carbendazim 25% + Flusilazole 12.5% SE | Scab (trial: 4 sprays at 15-day interval from first week of May) | 0.08% (best treatment in HP trial; 100 % control) | 10 L per 10 trees as in trial | V2 | YSPUHF Nauni trial, Kullu (VER). CIB&RC 2025 registers this mixture on apple for Alternaria blotch and premature leaf fall, not for scab |

Published ICAR schedule (NHB, from 1996; per 100 L water): silver tip to green tip - mancozeb 400 g or captan 300 g; pink bud - 'Contaf' 30 ml or 'Baycor' 50 g; petal fall - 'Bavistin' 50 g or 'Topsin M' 50 g; pea stage - mancozeb 300 g or captan 300 g; fruit development - 'Bavistin' + mancozeb (25 + 250 g); 15-20 days before harvest - mancozeb 300 g; before leaf fall - urea 5 kg (NHB). The source uses trade names. This is an older schedule: use the CIB&RC / label dose and PHI where they differ. Cross-check: bitertanol (Baycor), fenarimol and flusilazole 40% EC were listed for apple scab in the 2012 list but were not found for apple in the 2025 list, so they are not recommended here. In the HP trial the triazoles (flusilazole, difenoconazole, hexaconazole) gave 94-100 % control of scab on fruits and leaves, against about 67 % for carbendazim alone (VER). Several combination products have shorter PHIs (7-10 days) than single protectants; check PHI before the last pre-harvest spray.

## 6. Sources and citations


**[NHB]** National Horticulture Board (Govt. of India): Apple - Diseases (scab, fire blight, powdery mildew; spray schedule cited from 'ICAR: 50 years of crop science research in India', 1996). https://nhb.gov.in/pdf/fruits/apple/app002.pdf

**[VER]** Verma S., Gautam H.R. & Khosla K. (2022). Apple scab (Venturia inaequalis) management using a novel fungicide combination in the north-western Himalayas of India. International Journal of Economic Plants 9(1): 11-17 (Dr YS Parmar UHF, Nauni). https://ojs.pphouse.org/index.php/IJEP/article/download/4705/2161/4344

**[MIL]** Thakur V.S. & Khosla K. Relevance of Mills infection periods to apple scab prediction and rescheduling fungicide applications in Himachal Pradesh. Indian Journal of Agricultural Sciences (ICAR). https://epubs.icar.org.in/index.php/IJAgS/article/view/26355

**[FOR]** Studies on apple scab forecasting in Himachal Pradesh. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/20748

**[GK]** Greater Kashmir (press report): Scab outbreak hits South Kashmir apple orchards after incessant rains (supplementary, secondary source). https://m.greaterkashmir.com/article/scab-outbreak-hits-south-kashmir-apple-orchards-after-incessant-rains-growers-fear-heavy-losses/426490

**[BLO]** Epidemic outbreak of apple blotch disease: epidemiology and management in Himachal Pradesh. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/11109

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

**[CIB12]** CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for comparison only). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

# 2. Marssonina Leaf Blotch (Premature Leaf Fall)

Apple | Marssonina coronaria (sexual stage Diplocarpon mali) | Fungal (leaf-litter-borne)

**At a glance.** Dark brown blotches on the upper leaf surface from mid-June, leading to yellowing and early leaf fall. Prolonged warm, humid weather drives it. Preventive sprays and leaf-litter management are the focus.

## 1. Identification

### Symptoms

- Dark brown blotches on the upper side of the leaf, appearing from about the second week of June under humid weather (BLO).
- Starts as dark green circular patches on the upper surface that become 5-10 mm brown spots, darkening with time (PLX; supplementary source).
- Yellowing and premature defoliation follow (BLO). The disease is part of an 'apple blotch complex' of Marssonina and Alternaria species recorded in all apple-growing states (BLO).

### Distinguishing features

- Scab lesions are olive-green spots (NHB); blotch lesions are dark brown and larger, on the upper surface, and are followed by yellowing and leaf fall (BLO).

## 2. Causes and progression

- Pathogen: Marssonina coronaria, confirmed from isolates collected in Himachal Pradesh and Uttarakhand; conidia are hyaline, ampule-shaped, one-septate (UH2).
- Survival: in infected leaf litter. Acervuli were found in all leaf samples; conidia formation started in or after May; the ascosporic stage and apothecia were not observed in the Nauni samples (UH1).
- Spread: conidia released with rain. Maximum spore catch occurred on 7 August 2014 at 19 C, with ample rain and relative humidity above 85 % in the previous 24 h (UH1).
- Incubation: about 40 days from infection to obvious symptoms (PLX; supplementary).
- Leaf litter of Golden Delicious produced the most conidia, followed by Royal Delicious and Red Gold (UH1).

## 3. Favourable conditions

- Leaf ageing followed by mean temperature above 20 C and at least 70 % relative humidity continuously for 5-6 days induces yellowing and defoliation (BLO).
- Rainfall and relative humidity were positively correlated with spore release, and maximum temperature negatively (UH1).
- Older leaves are more susceptible than younger ones (PLX; supplementary).

## 4. Prevention and management

### Cultural practices, sanitation, resistant varieties and preventive measures

- Destroy fallen leaves in autumn to reduce spore carry-over (PLX; supplementary), consistent with the finding that the fungus survives in leaf litter (UH1).
- Use fungicides preventively rather than curatively (PLX; BLO tested them as preventive sprays).
- No apple cultivar is resistant; Granny Smith and Gibson's Golden are rated moderately resistant (PLX; supplementary).

### Regional note (Himalayan apple belt)

Described as one of the most destructive diseases wherever apple is grown in Himachal Pradesh, and recorded in other apple-growing states including Jammu & Kashmir (UH2; krishikosh thesis index and BLO).

## 5. Remedies and treatment

### Chemical options with verification status

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Kresoxim-methyl 44.3% SC | Apple blotch, leaf scab, fruit scab | 0.05% formulation (50 ml/100 L) | per tree size | V1 | CIB&RC 2025; PHI 7 d |
| Mancozeb 75% WG | Premature leaf fall, scab, Alternaria | 25-30 g per tree | 10 L/tree | V1 | CIB&RC 2025; PHI 40 d |

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| | spot/blight | | | | |
| Dodine 40% SC | Premature leaf fall; Alternaria leaf blight/blotch | 0.075% formulation | 10 L/tree | V1 | CIB&RC 2025; PHI 21 d |
| Tebuconazole 38.39% w/w SC | Marssonina leaf blotch (premature leaf fall) | 7.5 ml per 10 L water | per tree size | V1 | CIB&RC 2025; PHI 30 d |
| Difenoconazole 4% + Propineb 64% DF | Premature leaf fall (Marssonina coronaria), scab, powdery mildew | 0.125% formulation (125 g/100 L) | max 1000 L/ha | V1 | CIB&RC 2025; PHI 24 d |
| Fluopyram 17.7% + Tebuconazole 17.7% SC | Marssonina leaf blotch, Alternaria leaf spot, fruit rot | 0.5-0.63 ml/L | per tree size | V1 | CIB&RC 2025; PHI 30 d |
| Fluopyram 21.37% + Trifloxystrobin 21.37% SC | Marssonina blotch, scab, powdery mildew | 50-60 ml per 100 L | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Mancozeb 52.6% + Hexaconazole 2.4% WG | Powdery mildew and Marssonina blotch | 0.20% formulation | per canopy | V1 | CIB&RC 2025; PHI 20 d |
| Metiram 55% + Pyraclostrobin 5% WG | Premature leaf fall; Alternaria leaf spot and blight | 1750 g/ha (100 g/100 L) | per tree size | V1 | CIB&RC 2025; PHI 12 d |
| Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l SC | Premature leaf fall & Alternaria leaf spot | 15 ml per 100 L (187.5 g a.i./ha) | 2500 L/ha | V1 | CIB&RC 2025; PHI 29 d |
| Carbendazim 25% + Flusilazole 12.5% SE | Alternaria blotch and premature leaf fall | 0.08% formulation (0.03% a.i.) | 650-1000 L | V1 | CIB&RC 2025; PHI 9 d |
| Chlorothalonil 40% + Difenoconazole 4% SC | Powdery mildew, premature leaf fall, scab | 800 (ml/g) per ha | 1000 L | V1 | CIB&RC 2025; PHI 24 d |
| Hexaconazole 4% + Zineb 68% WP | Premature leaf fall, Alternaria leaf spot, scab, powdery mildew | 0.25% (25 g per 10 L) | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Azoxystrobin 11% + Tebuconazole 18.3% SC | Scab, powdery mildew & premature leaf fall | 1.0 (per litre, as printed) | 8-12 L/tree | V1 | CIB&RC 2025; PHI 10 d |
| Copper hydroxide ('Kocide') | Premature leaf fall (preventive) | 0.3% | - | V2 | ICAR journal trial, HP (BLO): 100 % control. No apple claim for copper hydroxide in the CIB&RC 2025 list |

The CIB&RC 2025 list now carries several apple claims for 'premature leaf fall' and 'Marssonina' (rows tagged V1), which is a change from the 2012 list. The HP Indian Phytopathology trial (BLO) separately reported 100 % preventive control with mancozeb 0.3%, propineb 0.3%, zineb 0.3%, carbendazim 0.05% and copper hydroxide 0.3% (product names printed in the abstract; spray number and timing not given). Only copper hydroxide remains outside the CIB&RC apple claims and is kept as a trial result (V2). Where a V1 claim reads 'premature leaf fall' it is the disease name used by CIB&RC; confirm on the product label which pathogen is claimed. Check the PHI of each product before the last pre-harvest spray.

## 6. Sources and citations


**[BLO]** Epidemic outbreak of apple blotch disease: epidemiology and management in Himachal Pradesh. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/11109

**[UH1]** Studies on the perpetuation of Diplocarpon mali causing premature leaf fall of apple and its management. Thesis, Dr YS Parmar UHF, Nauni (2014). https://krishikosh.egranth.ac.in/items/870cae0e-06e2-411c-aa72-b080abfb8ef1

**[UH2]** Studies on Marssonina blotch pathogenesis in apple and its management through novel methods. Thesis, Dr YS Parmar UHF, Nauni (2015). https://krishikosh.egranth.ac.in/handle/1/66655

**[PLX]** Plantix: Marssonina blotch of apple (company plant-health library; supplementary, non-university source). https://www.plantix.net/en/library/plant-diseases/100341/marssonina-blotch-of-apple/

**[NHB]** National Horticulture Board (Govt. of India): Apple - Diseases (scab, fire blight, powdery mildew; spray schedule cited from 'ICAR: 50 years of crop science research in India', 1996). https://nhb.gov.in/pdf/fruits/apple/app002.pdf

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

**[CIB12]** CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for comparison only). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

# 3. Powdery Mildew

Apple | Podosphaera leucotricha | Fungal (obligate, bud-borne)

**At a glance.** White powdery coating on shoots, leaves and blossoms, carried over in infected buds. Needs high humidity but not free water. Worst in nurseries and on young trees. Pruning out infected shoots plus sulphur-type sprays are the main tools.

## 1. Identification

**Symptoms**

- White, powdery (ash-like) coating in patches on both sides of leaves and on young shoots (NHB).
- Affected leaves turn pale and curl up; shoots stay weak and immature; severe infection causes leaf fall and premature fruit drop (NHB).
- Young infected fruit show russeting (NHB). Infected flowers appear covered with a powdery growth (OMA).

**Distinguishing features**

- Unlike scab, the symptom is a white powdery coating, and spore germination does not need free water (OSU).
- Nursery plants are more prone than orchard trees (NHB).

## 2. Causes and progression

- Pathogen: Podosphaera leucotricha; it needs living host tissue to survive (obligate fungus) (OMA).
- Survival: mycelium in dormant buds infected during the previous season (OSU, OMA).
- Progression: as buds break, the fungus colonises developing shoots (primary infection); wind-borne conidia then infect young leaves, flowers and fruit (secondary infection). Symptoms can be visible about 48 h after infection and a new crop of spores forms about 5 days later (OSU).

## 3. Favourable conditions

- Infection occurs at relative humidity above 90 % and about 10-25 C; the optimum is about 19-22 C (OSU gives 66-72 F; AGA gives 10-25 C and 19-22 C). Leaf wetness is not conducive, because spores do not germinate when immersed in water (OSU).
- Infections often occur in hot, humid weather when trees are growing rapidly (OMA).
- NHB describes the disease as appearing in dry climatic conditions, so it can occur in drier orchard weather where humidity around the canopy is high.
- Temperatures below about -24 C kill most infected buds (OMA); this is not a typical constraint in most Indian apple belts.

## 4. Prevention and management

**Cultural practices, sanitation, resistant varieties and preventive measures**

- Prune and destroy affected plant parts; in the field, spray after pruning out affected parts (NHB).
- Remove infected shoots as they appear in spring (OMA).
- Prune to open the canopy, reduce humidity and increase aeration and light (AGA).
- Cultivars differ greatly in susceptibility (OSU, OMA).
- Take extra care in nurseries and young plantings, where foliage infection is most damaging (NHB).

**Regional note (Himalayan apple belt)**

NHB describes it as a serious disease of buds, new shoots and leaves of apple in India; no state-wise incidence data were found in the sources reviewed.

## 5. Remedies and treatment

**Chemical options with verification status**

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Sulphur 80% WP | Powdery mildew | 2.5-5.0 kg/ha | 750-1000 L | V1 | CIB&RC 2025; PHI not stated |
| Lime sulphur 22% SC | Scale & powdery mildew | 2% before blossom and 1% after blossom | - | V1 | CIB&RC 2025; |
| Aureofungin 46.15% | Powdery mildew | 0.005% | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| W.V. SP | | | | | |
| Bupirimate 26.7% w/w EC | Powdery mildew | 1.5 ml/L | 8-12 L/tree | V1 | CIB&RC 2025; PHI 15 d |
| Metrafenone 500 g/l SC | Powdery mildew | 0.15 ml/L (187.5 g a.i./ha) | 2500 L/ha | V1 | CIB&RC 2025; PHI 42 d |
| Boscalid 25.2% + Pyraclostrobin 12.8% WG | Powdery mildew | 750 g/ha (285 g a.i.) | 2500 L/ha | V1 | CIB&RC 2025; PHI 41 d |
| Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l SC | Premature leaf fall, Alternaria leaf spot and powdery mildew | 25 ml per 100 L (12.5 g a.i./100 L) | per tree size | V1 | CIB&RC 2025; PHI 41 d |
| Fluxapyroxad 75 g/l + Difenoconazole 50 g/l SC | Scab and powdery mildew | 1000 ml/ha | 2500 L | V1 | CIB&RC 2025; PHI 33 d |
| Chlorothalonil 40% + Difenoconazole 4% SC | Powdery mildew, premature leaf fall, scab | 800 (ml/g) per ha | 1000 L | V1 | CIB&RC 2025; PHI 24 d |
| Difenoconazole 4% + Propineb 64% DF | Scab, powdery mildew, premature leaf fall | 0.125% formulation (125 g/100 L) | max 1000 L/ha | V1 | CIB&RC 2025; PHI 24 d |
| Fluopyram 21.37% + Trifloxystrobin 21.37% SC | Marssonina blotch, scab, powdery mildew | 50-60 ml per 100 L | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Mancozeb 52.6% + Hexaconazole 2.4% WG | Powdery mildew and Marssonina blotch | 0.20% formulation | per canopy | V1 | CIB&RC 2025; PHI 20 d |
| Carbendazim 12% + Mancozeb 63% WP | Fruit scab & powdery mildew | 0.25% formulation | per canopy | V1 | CIB&RC 2025; PHI 20 d |
| Hexaconazole 4% + Zineb 68% WP | Powdery mildew (also scab, Alternaria, core rot) | 0.25% (25 g per 10 L) | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Tebuconazole 6.7% + Captan 26.9% SC | Powdery mildew, Alternaria leaf spot/blight, scab | 0.25% (2.5 ml/L) | 10 L/tree | V1 | CIB&RC 2025; PHI 10 d |
| Azoxystrobin 11% + Tebuconazole 18.3% SC | Scab, powdery mildew & premature leaf fall | 1.0 (per litre, as printed) | 8-12 L/tree | V1 | CIB&RC 2025; PHI 10 d |
| Sulphur | Powdery mildew | 0.3% | - | V2 | NHB (ICAR schedule) |
| Carbendazim | Powdery mildew | 0.05% | - | V2 | NHB (ICAR schedule); CIB&RC lists carbendazim on apple for scab only |
| Triadimefon ('Bayleton') | Powdery mildew in nursery seedlings | 500 ppm at 7-day interval | - | V2 | NHB (ICAR schedule) |

Cross-check: dinocap 48% EC (apple powdery mildew) was in the 2012 list but the 2025 list shows dinocap only for mango and rose, so it is no longer recommended here; NHB's 'Karathane' (dinocap) 0.05% is therefore not reproduced. Sulphur can burn foliage in hot weather and some fruit varieties are sensitive; read the label. Many scab fungicides also suppress powdery mildew, but not all are highly effective on it, and powdery mildew often occurs when scab infection periods are unlikely (OMA). Several powdery-mildew entries have long PHIs (30-42 days): plan sprays accordingly.

## 6. Sources and citations


**[NHB]** National Horticulture Board (Govt. of India): Apple - Diseases (scab, fire blight, powdery mildew; spray schedule cited from 'ICAR: 50 years of crop science research in India', 1996). https://nhb.gov.in/pdf/fruits/apple/app002.pdf

**[OSU]** Ohio State University Extension (CFAES): Apple powdery mildew, fact sheet PLPATH-FRU-21 (biology and weather; used for biology only). https://ohioline.osu.edu/factsheet/plpath-fru-21

**[OMA]** OMAFRA (Ontario Ministry of Agriculture): Apple powdery mildew (biology and scouting; used for biology only). https://www.omafra.gov.on.ca/IPM/english/apples/diseases-and-disorders/powdery-mildew.html

**[AGA]** AgroAtlas: Podosphaera leucotricha - powdery mildew of apple (infection temperature and humidity). https://agroatlas.ru/en/content/diseases/Pomae/Pomae_Podosphaera_leucotricha

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register):

https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

**[CIB12]** CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for comparison only). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

# 4. Collar Rot

Apple | Phytophthora cactorum | Oomycete (soil-borne)

**At a glance.** Soil-borne rot at the soil line that can girdle and kill trees. Early diagnosis is hard. Site drainage, resistant rootstocks and soil-applied biocontrol / oil-cake options are the focus; no verified fungicide recommendation.

## 1. Identification

**Symptoms**

- The fungus attacks trees at soil level and can girdle the stem or crown completely, leading to death of the plant (COL).
- Infected trees may bloom normally but fruits are small, leaves wilt and drop, and the tree declines gradually; symptoms develop over several seasons (MSU).
- Infected bark shows a reddish-brown inner-bark discolouration below the soil line with a clear-cut margin between diseased and healthy tissue (MSU).

**Distinguishing features**

- Other root and crown problems give similar above-ground symptoms: rootstock blight (fire blight), 'wet feet' (root asphyxiation), borers, winter injury and apple union necrosis (MSU).
- White root rot shows a white mycelial mat on dark lateral roots (WR1), whereas collar rot is centred on the collar/bark at the soil line (COL, MSU).

## 2. Causes and progression

- Pathogen: Phytophthora cactorum, a soil-borne pathogen prevalent throughout apple-growing areas (COL).
- Incidence: in a survey of Mandi, Kullu and Shimla districts of Himachal Pradesh, the highest mean incidence was 34.5 % in Rohanda block of Mandi (CO2).

## 3. Favourable conditions

- The Indian sources reviewed give no temperature or humidity thresholds for collar rot.
- Extension guidance links the disease to wet sites and states that resistant rootstocks are necessary on historically wet sites (MSU).

## 4. Prevention and management

**Cultural practices, sanitation, resistant varieties and preventive measures**

- Plant on well-drained sites and use resistant rootstocks on historically wet sites (MSU).
- Soil solarization, non-host crops, biofumigation and oil cakes have been reported to help manage collar rot and white root rot (Sharma and Negi 2013, as cited in COL).
- Pot-culture trial (Nauni / Uttarakhand): mustard cake water extract applied 7 days BEFORE the pathogen gave 13.3 % disease incidence against about 85 % in the untreated control; eucalyptus leaf extract gave 13.3 % in the same timing (COL). These are pot results, not a field recommendation.
- Pot trial of soil-applied antagonists: Bacillus cereus gave 84.2 %, Exiguobacterium aquaticum 78.9 % and Trichoderma virens 76.3 % disease control (CO2). These are not registered products.

**Regional note (Himalayan apple belt)**

Described as one of the most important soil-borne diseases of apple, prevalent across apple-growing areas of Jammu & Kashmir, Himachal Pradesh and Uttarakhand (COL, CO2).

## 5. Remedies and treatment

**Chemical options with verification status**

No chemical recommendation is given (tag --).

No chemical recommendation is given (tag --). The Indian paper reviewed states that soil drenching with fungicides is not cost-effective, that chemicals often fail to reach the infection site, and that they can aggravate other soil-borne diseases (COL). The CIB&RC 2025 list shows no collar-rot or Phytophthora claim on apple. For a specific orchard, consult the local KVK / YSPUHF Nauni / SKUAST and verify any product on the CIB&RC register before use.

## 6. Sources and citations

**[COL]** Negi H.S., Sharma I.M. & Upadhyay R.G. (2017). Evaluation of botanicals against Phytophthora cactorum causing collar rot in apple. International Journal of Tropical Agriculture 35(4). https://serialsjournals.com/abstract/47598_32-.pdf

**[CO2]** Biological control of collar rot (Phytophthora cactorum) of apple. Thesis, Dr YS Parmar UHF, Nauni (2023). https://krishikosh.egranth.ac.in/items/d4310217-f003-4134-a922-bf5c5c285602

**[MSU]** Michigan State University Extension: Phytophthora root, crown, and collar rot (symptoms and look-alikes; used for biology only). https://www.canr.msu.edu/ipm/diseases/phytophthora_root_crown_and_collar_rot

**[WR1]** Kumar A., Attri B.L., Kishor A., Debnath S., Mer M.S. & Narayan R. (2020). Influence of rootstocks on white root rot (Dematophora necatrix) resistance in apple (Malus baccata). Indian Journal of Agricultural Sciences 90(1): 53-57 (ICAR-CITH, Regional Station Mukteshwar). https://epubs.icar.org.in/index.php/IJAgS/article/download/98530/39129/341426

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

# 5. White Root Rot

Apple | Dematophora necatrix (Rosellinia necatrix) | Fungal (soil-borne)

**At a glance.** A major and destructive root disease in nurseries and orchards. Symptoms show on the canopy only late, so detection is hard. Resistant rootstock (Paron), nursery-site rotation and soil treatments are the main options.

## 1. Identification

**Symptoms**

- Above ground: wilting is the first symptom, followed by bronzing and inward cupping of leaves and defoliation (WR1).
- Below ground: lateral roots turn dark brown and are covered with a white mycelial mat (WR1).
- Above-ground symptoms appear only at a very advanced stage; there is no quick early detection method (WR1).
- Seedlings are killed within about three weeks; adult trees survive two to three seasons (WR1).

**Distinguishing features**

- The pathogen's pear-shaped swelling at the hyphal septum is the characteristic microscopic feature (WR1, WR3).
- White mycelial mat on dark lateral roots separates it from collar rot, which centres on the bark at the soil line (WR1, COL).

## 2. Causes and progression

- Pathogen: Dematophora necatrix Hartig (sexual stage Rosellinia necatrix), a soil-borne fungus (WR1, WR3).
- In India it was first observed in the Uttar Pradesh hills (now Uttarakhand) in 1943 and is now reported from almost all apple-growing regions (WR1).
- Incidence in Nainital district, Uttarakhand: mean 52.9 % in nurseries against 28.0 % in orchards, and 70.3 % (nursery) and 38.3 % (orchard) at the worst location (WR1).
- Inoculum builds up where nursery sites are not rotated: one nursery not rotated for a decade lost 100 % of one-year-old grafted plants in earlier years (WR1).
- The fungus produces a thermostable toxin (WR1). In pathogenicity tests symptoms appeared 13-15 days after inoculation (WR3).

## 3. Favourable conditions

- The rainy season is favourable to development of the disease (WR1).
- Trees that are nutritionally weak are highly prone; in Kumaon hills the soils are mostly sandy with poor water-holding capacity and summer rainfall is scanty, which stresses shallow-rooted trees (WR1).

## 4. Prevention and management

**Cultural practices, sanitation, resistant varieties and preventive measures**

- Rootstock choice: Paron (Malus baccata var. himalaica) was resistant (disease severity 4.35 %, mortality 15.3 % at the highest inoculum dose); MM 111 (18.3 %) and M 9 (14.8 %) were moderately resistant; MM 106 (59.8 %) was susceptible and Srinagar crab (69.8 %) highly susceptible (WR1). Paron is registered with NBPGR (IC-0614799) (WR1). No rootstock is immune (WR1).
- Malus baccata from Kashmir and Shillong also showed tolerance in a germplasm screen (WR2).
- Rotate nursery sites; do not raise nursery plants repeatedly in infested soil (WR1).
- Cauliflower, cabbage and maize can be used as intercrops in orchards infested with D. necatrix (WR3).
- Soil solarization, non-host crops, biofumigation and oil cakes are reported to help against collar rot and white root rot (Sharma and Negi 2013, cited in COL).

**Regional note (Himalayan apple belt)**

A major soil-borne disease of apple in Jammu & Kashmir, Himachal Pradesh and Uttarakhand, affecting nurseries and orchards (WR1, WR3).

## 5. Remedies and treatment

### Chemical options with verification status

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Aureofungin 46.15% W.V. SP | White root rot | 0.2% solution; the CIB&RC entry lists '5-10 injection, 2 g/tree' (layout unclear). Follow the label | - | V1 | CIB&RC 2025; PHI 30 d |

The ICAR paper reports that carbendazim is extensively used against white root rot in India (WR1), but the CIB&RC 2025 list shows no white-root-rot claim for carbendazim on apple and the sources reviewed give no verified dose, so no carbendazim dose is included. The Aureofungin entry above is the only apple white-root-rot claim found in the CIB&RC 2025 list. The printed layout of the dose columns is unclear, so confirm method, concentration and quantity on the product label.

## 6. Sources and citations


**[WR1]** Kumar A., Attri B.L., Kishor A., Debnath S., Mer M.S. & Narayan R. (2020). Influence of rootstocks on white root rot (Dematophora necatrix) resistance in apple (Malus baccata). Indian Journal of Agricultural Sciences 90(1): 53-57 (ICAR-CITH, Regional Station Mukteshwar). https://epubs.icar.org.in/index.php/IJAgS/article/download/98530/39129/341426

**[WR2]** Sharma Y.P. et al. (2013). Disease reaction of apple germplasm to white root rot (Dematophora necatrix). Indian Journal of Horticulture 70(1): 130-134. https://journal.iahs.org.in/index.php/ijh/article/download/1467/743

**[WR3]** Studies on the integrated management of white root rot (Dematophora necatrix Hartig) in apple. Thesis, Dr YS Parmar UHF, Nauni (2018). https://krishikosh.egranth.ac.in/items/cc573cdc-314b-43e5-af6f-379765cc74b5

**[COL]** Negi H.S., Sharma I.M. & Upadhyay R.G. (2017). Evaluation of botanicals against Phytophthora cactorum causing collar rot in apple. International Journal of Tropical Agriculture 35(4). https://serialsjournals.com/abstract/47598_32-.pdf

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

# 6. Alternaria Leaf Blotch

Apple | Alternaria alternata f. sp. mali (Alternaria mali); other Alternaria spp. also associated | Fungal (leaf-litter-borne, toxin-producing)

**At a glance.** Reddish-brown leaf spots with purplish borders that enlarge into zonate blotches and cause early leaf fall. Often occurs with Marssonina blotch as an 'apple blotch complex'. Preventive sprays and leaf-litter management are used; no cultivar tested was resistant.

## 1. Identification

### Symptoms

- Reddish-brown necrotic spots with dark purplish borders on leaves; dark brown to black lesions on petioles and midribs, causing yellowing and defoliation (AL2).
- Leaf spots enlarge into zonate, circular or crescent-shaped rings; infection starts early in the season and leads to premature defoliation (AL4).
- The fungus infects leaves and fruits (AL4).

### Distinguishing features

- Marssonina blotch gives dark green circular patches that turn brown on the upper surface (PLX), whereas Alternaria spots are reddish-brown with purplish borders and can extend to petioles and midribs (AL2).
- The two diseases occur together as an 'apple blotch complex' (BLO), so laboratory confirmation may be needed to separate them.

## 2. Causes and progression

- Pathogen: Alternaria alternata f. sp. mali (A. mali). In Jammu & Kashmir, A. alternata, A. tenuissima and A. arborescens were all found associated with apple leaf disease (AL3).
- The pathogen produces a host-specific toxin: a leaf necrosis bioassay gave a positive reaction on the susceptible cultivar Royal Delicious (AL1).
- Symptoms appeared 4-7 days after inoculation in pathogenicity tests (AL1).
- Leaf litter acts as a source of primary inoculum: dipping leaf litter in urea (50,000 ppm) gave maximum suppression of primary inoculum in thesis studies (AL1).

## 3. Favourable conditions

- Occurs every season in wet climates, in moderate to severe form in Shimla district apple areas (AL4).
- For the blotch complex (Marssonina and Alternaria): leaf ageing followed by mean temperature above 20 C and at least 70 % relative humidity for 5-6 continuous days induces yellowing and defoliation (BLO).

## 4. Prevention and management

### Cultural practices, sanitation, resistant varieties and preventive measures

- Manage leaf litter: urea dip of leaf litter suppressed primary inoculum in laboratory/thesis studies (AL1). This is not a field recommendation.
- Of eight plant extracts tested in vitro, neem leaf extract was most effective (72.9 % inhibition) (AL1). In vitro result only.
- Host resistance: of 11 apple cultivars screened, none was moderately resistant or resistant (AL2).
- Use preventive rather than curative sprays (BLO tested sprays as preventives).

### Regional note (Himalayan apple belt)

An economically important disease of apple in Himachal Pradesh, first documented from Himachal Pradesh and Kashmir (AL1, AL3). Disease incidence and severity were highest in Jubbal, Shimla, in 2022-23 (AL2).

## 5. Remedies and treatment

### Chemical options with verification status

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Dodine 40% SC | Alternaria leaf blight/blotch; premature leaf fall | 0.075% formulation | 10 L/tree | V1 | CIB&RC 2025; PHI 21 d |

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Hexaconazole 75% WG | Alternaria leaf spot | 0.004% formulation (4 g/100 L) | per tree size | V1 | CIB&RC 2025; PHI 30 d |
| Mancozeb 75% WG | Scab, premature leaf fall, Alternaria spot/blight, sooty blotch | 25-30 g per tree | 10 L/tree | V1 | CIB&RC 2025; PHI 40 d |
| Carbendazim 25% + Flusilazole 12.5% SE | Alternaria blotch and premature leaf fall | 0.08% formulation (0.03% a.i.) | 650-1000 L | V1 | CIB&RC 2025; PHI 9 d |
| Metiram 55% + Pyraclostrobin 5% WG | Premature leaf fall; Alternaria leaf spot and blight | 1750 g/ha (100 g/100 L) | per tree size | V1 | CIB&RC 2025; PHI 12 d |
| Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l SC | Premature leaf fall & Alternaria leaf spot | 15 ml per 100 L (187.5 g a.i./ha) | 2500 L/ha | V1 | CIB&RC 2025; PHI 29 d (41 d for the 25 ml/100 L entry that also covers powdery mildew) |
| Captan 70% + Hexaconazole 5% WP | Scab and Alternaria blotch | 0.06-0.1% (6-10 g per 10 L) | per tree size | V1 | CIB&RC 2025; PHI 10 d |
| Fluopyram 17.7% + Tebuconazole 17.7% SC | Marssonina leaf blotch, Alternaria leaf spot, fruit rot | 0.5-0.63 ml/L | per tree size | V1 | CIB&RC 2025; PHI 30 d |
| Hexaconazole 4% + Zineb 68% WP | Premature leaf fall, Alternaria leaf spot/blight, scab, powdery mildew, core rot | 0.25% (25 g per 10 L) | 10 L/tree | V1 | CIB&RC 2025; PHI 30 d |
| Tebuconazole 6.7% + Captan 26.9% SC | Powdery mildew, Alternaria leaf spot/blight, scab | 0.25% (2.5 ml/L) | 10 L/tree | V1 | CIB&RC 2025; PHI 10 d |

These V1 rows replace the earlier trial-only table: the CIB&RC 2025 list carries specific apple claims for Alternaria leaf blotch/spot/blight and premature leaf fall. The earlier HP trial (BLO) on the combined Marssonina-plus-Alternaria complex (mancozeb 0.3%, propineb 0.3%, zineb 0.3%, carbendazim 0.05%, copper hydroxide 0.3%) is not a label claim for those products. Check the label for the exact pathogen claim, number of sprays and PHI. Dr YS Parmar UHF Nauni and the HP Department of Horticulture ran a joint awareness campaign on Alternaria and Marssonina blotch in Feb 2026; contact them for current orchard advice.

## 6. Sources and citations


**[AL1]** Studies on Alternaria blotch of apple and its management. Thesis, Dr YS Parmar UHF, Nauni (2018). https://krishikosh.egranth.ac.in/items/ea9160b1-29d6-4edf-aa79-4265986c8905

**[AL2]** Occurrence, etiology and epidemiology of Alternaria leaf blotch of apple in Himachal Pradesh. Thesis, Dr YS Parmar UHF, Nauni (2023). https://krishikosh.egranth.ac.in/handle/1/5810207107

**[AL3]** Multiple associations of Alternaria species on high density apple plantation system in Jammu and Kashmir, India. Indian Journal of Agricultural Sciences 95(8): 989-995 (2025). https://epubs.icar.org.in/index.php/IJAgS/article/view/155250

**[AL4]** Disease status of Alternaria blotch of apple in Himachal Pradesh, India. Indian Journal of Agricultural Sciences. https://i-scholar.in/index.php/Ijas/article/view/180170

**[BLO]** Epidemic outbreak of apple blotch disease: epidemiology and management in Himachal Pradesh. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/11109

**[PLX]** Plantix: Marssonina blotch of apple (company plant-health library; supplementary, non-university source). https://www.plantix.net/en/library/plant-diseases/100341/marssonina-blotch-of-apple/

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

**[CIB12]** CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for comparison only). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

# 7. Sooty Blotch and Flyspeck

Apple | Fungal complex: Peltaster fructicola, Geastrumia polystigmatus, Leptodontium elatus (sooty blotch); Zygophiala jamaicensis (flyspeck) | Fungal (surface blemish; twig-borne)

**At a glance.** Sooty smudges and clusters of black specks on the fruit surface in humid late-season weather. Fruit is not rotted but is downgraded for appearance. Canopy aeration and scab-programme fungicides reduce it.

## 1. Identification

**Symptoms**

- Sooty blotch: sooty smudges or olive-green spots on mature fruit (SB1). Flyspeck: clusters of about 10-50 sharply defined, black, shiny specks on the fruit surface (SB1).
- The fungi do not cause decay; losses come from unacceptable appearance, though fruit may lose water faster and store less well (SB1).
- Symptoms are more apparent on yellow, green or light-coloured fruit (SB1).

**Distinguishing features**

- Unlike scab, there is no corky or cracked lesion and no leaf symptom; the blemish is superficial (SB1, SB2).
- Sooty blotch appears only on fruit, whereas sooty mould can also grow on leaves and twigs and wipes off more easily (SB2).

## 2. Causes and progression

- Many fungi are involved, including Peltaster fructicola, Geastrumia polystigmatus, Leptodontium elatus and Zygophiala jamaicensis (SB1).
- The fungi are commonly found on the stem surfaces of many woody plants, including apple shoots (SB1); they overwinter on twigs of apple, pear and other hosts (UW).
- The fungi grow slowly, so symptoms appear mostly during the harvest season even though infection may have happened much earlier (SB2).

## 3. Favourable conditions

- Mycelial growth that forms sooty blotches can occur without free water at relative humidity above 90 % (SB1).
- Optimum conditions for flyspeck conidial production: 60-70 F (about 16-21 C) and relative humidity above 96 % (SB1).
- Laboratory study: P. fructicola conidia germinated at 12-24 C and RH of 95 % or more; mycelial growth was inhibited below 95 % RH and absent at 88 % RH (SB3).
- These data are from US/Canadian studies; no Indian field study on these diseases was found in this review.

## 4. Prevention and management

**Cultural practices, sanitation, resistant varieties and preventive measures**

- Use cultural practices that dry the tree and fruit, for example pruning and canopy management to improve airflow and reduce humidity at the fruit (SB2, UKY).
- Fungicide programmes aimed at flyspeck also provide later control of sooty blotch (SB2).
- Be aware that reservoir hosts such as other woody plants near the orchard can carry the fungi (UKY).

**Regional note (Himalayan apple belt)**

No Indian incidence or state data were found for this disease in the sources reviewed; the biology above is from non-Indian extension sources and is a guide only.

## 5. Remedies and treatment

**Chemical options with verification status**

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Mancozeb 75% WP | Scab & sooty blotch | 30 g per tree in 10 L water | 10 L/tree | V1 | CIB&RC 2025; PHI not stated |
| Mancozeb 75% WG | Scab, premature | 25-30 g per tree | 10 L/tree | V1 | CIB&RC 2025; PHI 40 d |

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| | leaf fall, Alternaria, sooty blotch | | | | |
| Captan 75% WP | Scab, fly speck, bitter rot | 0.12% a.i. (1667 g formulation per 1000 L) | 15-20 L/tree | V1 | CIB&RC 2025; PHI 8 d. Footnote warns of possible 'resetting' (as printed; probably russeting) on Red Delicious |

These are the only apple claims for sooty blotch or fly speck found in the 2025 CIB&RC list. They overlap with the scab programme, so timing follows the scab schedule and the label. Confirm the current registration and the PHI before use.

## 6. Sources and citations


**[SB1]** Sooty blotch and flyspeck in apple. eXtension apples (U.S. land-grant extension; used for biology only). https://apples.extension.org/sooty-blotch-and-flyspeck-in-apple

**[SB2]** OMAFRA (Ontario): Sooty blotch (biology and management notes; used for biology only). https://www.omafra.gov.on.ca/IPM/english/apples/diseases-and-disorders/sooty-blotch.html

**[SB3]** Effects of temperature and relative humidity on Peltaster fructicola and Leptodontium elatius, causes of apple sooty blotch. Phytopathology 90(4): 362 (NC State University). https://apsnet.org/publications/phytopathology/2000/April/Pages/90_4_362.aspx

**[UW]** University of Wisconsin-Extension A3173: Apple and pear disorder - sooty blotch and flyspeck (used for biology only). https://hort.extension.wisc.edu/files/2015/02/A3173.pdf

**[UKY]** University of Kentucky Extension: Sooty blotch and flyspeck of apple, PPFS-FR-T-16 (used for biology only). https://plantpathology.ca.uky.edu/files/ppfs-fr-t-16.pdf

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# 8. Fire Blight

Apple | Erwinia amylovora | Bacterial (blossom- and shoot-infecting)

**At a glance.** Bacterial disease that scorches shoots and blossoms and can spread rapidly in spring. NHB advises burning affected trees and host plants and spraying streptomycin to protect spring blossom. The CIB&RC list carries an apple fire blight claim for streptomycin + tetracycline.

## 1. Identification

### Symptoms

- Distinct fire-like (scorched) appearance on infected plant parts; new shoots are highly susceptible (NHB).
- Shoot tips wilt and droop without browning; golden-coloured bacterial ooze is seen on the stem (NHB).
- Fruit: necrotic spots and oozing lesions on the outer surface (NHB).

### Distinguishing features

- Wilted shoot tips that droop without browning, together with golden bacterial ooze, separate fire blight from scab and the fungal blotches, which cause leaf spots (NHB).
- Several root and crown problems also give wilting above ground; check for the ooze and the fire-like scorch on shoots (NHB; MSU lists rootstock blight as a look-alike of collar rot).

## 2. Causes and progression

- Pathogen: Erwinia amylovora, a bacterium (NHB).
- The NHB text describes spring blossom infection; no detail on survival, spread or incidence in Indian orchards was given in the sources reviewed.

## 3. Favourable conditions

- The sources reviewed give no temperature, humidity or rainfall thresholds for fire blight in India. NHB links infection to spring blossom (NHB).

## 4. Prevention and management

### Cultural practices, sanitation, resistant varieties and preventive measures

- Collect and burn affected trees and host plants immediately when fire blight is seen (NHB).
- Sprays of streptomycin protect spring blossom from infection (NHB).
- CIB&RC instructs spraying at 20-30 % bloom, repeated every 3-4 days during blossom time (CIB).

### Regional note (Himalayan apple belt)

NHB lists fire blight among the diseases of apple in India; the sources reviewed give no state-wise incidence. Confirm local status with SKUAST-Kashmir, YSPUHF Nauni or the state horticulture department before relying on this entry.

## 5. Remedies and treatment

### Chemical options with verification status

| Product | Use / target | Dose (formulation) | Water | Tag | Basis / PHI |
|---|---|---|---|---|---|
| Streptomycin sulphate 90% + Tetracycline hydrochloride 10% SP ('Streptocycline') | Apple fire blight | 25-50 ppm solution at 20-30 % bloom; repeat every 3-4 days during blossom | - | V1 | CIB&RC 2025; waiting period not stated |
| Streptomycin sulphate 9% + Tetracycline hydrochloride 1% SP | Apple fire blight | 25-50 ppm solution at 20-30 % bloom; repeat every 3-4 days during blossom | - | V1 | CIB&RC 2025; waiting period not stated |

The antibiotic dose is given as a solution strength in ppm, not per hectare. Use only as the label directs, mind the limits on repeated antibiotic sprays, and follow local horticulture department advice for your area. Sanitation (burning affected trees and host plants) comes first (NHB).

## 6. Sources and citations

**[NHB]** National Horticulture Board (Govt. of India): Apple - Diseases (scab, fire blight, powdery mildew; spray schedule cited from 'ICAR: 50 years of crop science research in India', 1996). https://nhb.gov.in/pdf/fruits/apple/app002.pdf

**[MSU]** Michigan State University Extension: Phytophthora root, crown, and collar rot (symptoms and look-alikes; used for biology only). https://www.canr.msu.edu/ipm/diseases/phytophthora_root_crown_and_collar_rot

**[CIB]** Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

# 9. Cedar Apple Rust

Apple | Gymnosporangium juniperi-virginianae | Fungal rust (needs two hosts: juniper and apple)

**At a glance.** A rust that must alternate between juniper (eastern red cedar) and apple. It is a major disease where junipers grow near apple orchards in North America. No Indian report or Indian-labelled recommendation was found, so this chapter is mainly diagnostic and preventive information.

## 1. Identification

**Symptoms**

- On apple: the most conspicuous symptom is bright orange lesions on the leaves (CAR1); lesions on fruit start as orange, slightly raised spots (CAR2).
- Aecia (finger-like structures) form on the underside of apple leaves below the orange spots (CAR2).
- On juniper / eastern red cedar: brown, globular galls (1/4 to 2 inches) that produce bright orange gelatinous tendrils (telial horns) after spring rain (CAR1, CAR2).

**Distinguishing features**

- Scab gives olive-green spots (NHB); cedar apple rust gives bright orange to yellow spots on the upper leaf surface (CAR1, CAR4).
- Cedar-quince and cedar-hawthorn rusts are related species with similar life cycles that differ in the juniper hosts and the symptoms they cause (CAR4).

## 2. Causes and progression

- Pathogen: Gymnosporangium juniperi-virginianae; it needs both a rosaceous host (apple, crabapple) and a juniper such as eastern red cedar (Juniperus virginiana) to complete its life cycle (CAR1).
- The full cycle takes 24 months and requires infection of two different hosts (CAR3). Spores released from the juniper galls infect apple in spring; spores from the apple leaf spots infect juniper from midsummer into early fall (CAR3).
- Spores are carried by wind; the spore stage produced on apple cannot reinfect apple (CAR1).

## 3. Favourable conditions

- Spore release from juniper galls follows spring rain, so wet spring weather during early apple growth stages is critical (CAR2, CAR1).
- Risk is tied to the distance to juniper: in the US, apple trees within about 2-5 miles of the alternate host should be managed annually from tight cluster to about 14 days after petal fall (CAR2); Illinois Extension reports that removing junipers within 3 miles of an orchard makes risk minimal to none (CAR3).

## 4. Prevention and management

**Cultural practices, sanitation, resistant varieties and preventive measures**

- Avoid growing apple near susceptible junipers; remove nearby juniper hosts where practical (CAR1, CAR4).
- Remove galls from junipers in early spring before the telial horns form (CAR3).
- Plant resistant varieties where available (CAR4).

**Regional note (Himalayan apple belt)**

No Indian report of cedar apple rust on apple was found in the sources reviewed. The principal juniper host named in the literature, eastern red cedar, is a North American tree; the disease needs a susceptible juniper near the orchard. Treat this chapter as background to recognise the disease, and confirm any suspected case with a plant pathologist.

## 5. Remedies and treatment

**Chemical options with verification status**

No chemical recommendation is given (tag --).

No chemical recommendation is given (tag --). The CIB&RC 2025 fungicide list has no apple rust claim, and the Indian sources reviewed give no recommendation. US extension notes that sterol-inhibiting (FRAC 3) fungicides are most effective and that scab sprays have historically controlled the rust (CAR2), but those are US labels and are not Indian recommendations. If rust is confirmed in an orchard, consult SKUAST-Kashmir, YSPUHF Nauni or the state horticulture department.

## 6. Sources and citations

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**[CAR2]** NC State Extension: Cedar apple rust (life cycle, timing and fungicide groups in the US; used for biology only). https://content.ces.ncsu.edu/cedar-apple-rusts

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**[CAR4]** Purdue Extension BP-138-W: Cedar apple and related rusts (Gymnosporangium species; used for biology only). https://www.extension.purdue.edu/extmedia/BP/BP-138-W.pdf

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**[BLO]** Epidemic outbreak of apple blotch disease: epidemiology and management in Himachal Pradesh. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/11109

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## Items checked but not included

- Diseases not covered in this nine-disease set: apple canker (CIB&RC 2025 lists copper sulphate + mancozeb for apple canker; no biology source retrieved), post-harvest fruit rots (CIB&RC lists fludioxonil for apple Botrytis, Penicillium, Alternaria and Mucor rots), bitter rot, apple mosaic virus and replant problems.
- State-wise spray calendars from SKUAST-Kashmir, YSPUHF Nauni or the HP Horticulture Department: none were retrieved, so only the ICAR schedule published by NHB (1996) is shown, with a caution that it is dated.
- Collar rot: no apple Phytophthora claim in the CIB&RC 2025 list, so chemical advice is limited to cultural and biological information. The Marssonina/Alternaria trial-only table was replaced by CIB&RC V1 rows after cross-checking the 2025 list; copper hydroxide stays V2.
- Cross-check of the 2012 list against 2025: bitertanol, fenarimol, flusilazole 40% EC (apple scab) and dinocap (apple powdery mildew) were not found for apple in 2025 and were removed. Registration changes after 30/09/2025 were not checked.
- The 2025 compilation was read from a copy on a university website, not the live CIB&RC register.